

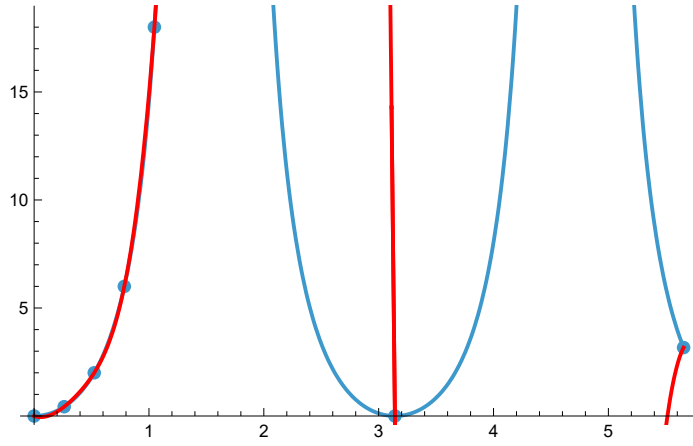
Вариант 3;

Задание 1

```
Clear[P, PP, PPP, A, B, eq, koef, T, X, F, NN, n, xx, spl1, spl2, spl3];
f[x_] = 6 Tan[x]^2;
X = {0, Pi/6, Pi/4, Pi/3, Pi/12, Pi, 9 Pi/5}; xx = Pi/9;
F = f[X];
Subscript[x, k_] := X[[k + 1]];
Subscript[f, k_] := F[[k + 1]];
NN = Length[X];
n = (NN - 1)/2;
koef = Solve[Table[Subscript[a, 0] + Sum[Subscript[a, k] Sin[k*Subscript[x, j]] + Subscrip
T[x_] = Subscript[a, 0] + Sum[Subscript[a, k] Sin[k*x] + Subscript[b, k] Cos[k*x], {k, 1,
Table[PossibleZeroQ[T[Subscript[x, j]] - Subscript[f, j]], {j, 0, NN - 1}]
Gr1 = ListPlot[MapThread[List, {X, F}], PlotStyle -> {PointSize[0.02]}];
Gr2 = Plot[f[y], {y, Subscript[x, 0], Subscript[x, NN - 1]}];
Gr3 = Plot[T[y], {y, Subscript[x, 0], Subscript[x, NN - 1]}, PlotStyle -> Red];
Show[Gr1, Gr2, Gr3]
T[xx] // N
Abs[f[xx] - T[xx]]
```

$$\begin{aligned}
& \left(2 \sin[x] \left(-91950 - 70935 \sqrt{2} + 55830 \sqrt{3} + 28878 \sqrt{5} + 39015 \sqrt{6} + 26901 \sqrt{10} - 18390 \sqrt{15} - \right. \right. \\
& \quad 14317 \sqrt{30} - 18644 \sqrt{30 - 6 \sqrt{5}} - 22146 \sqrt{15 - 3 \sqrt{5}} + 32496 \sqrt{10 - 2 \sqrt{5}} + 38070 \sqrt{5 - \sqrt{5}} - \\
& \quad 48864 \sqrt{5 + \sqrt{5}} - 42330 \sqrt{2(5 + \sqrt{5})} + 28632 \sqrt{3(5 + \sqrt{5})} + 24142 \sqrt{6(5 + \sqrt{5})} + \\
& \quad 2 \left(79410 + 53595 \sqrt{2} - 45180 \sqrt{3} - 32760 \sqrt{5} - 31415 \sqrt{6} - 17757 \sqrt{10} + 18774 \sqrt{15} + \right. \\
& \quad \quad 10351 \sqrt{30} + 7443 \sqrt{30 - 6 \sqrt{5}} + 14400 \sqrt{15 - 3 \sqrt{5}} - 12501 \sqrt{10 - 2 \sqrt{5}} - 25494 \sqrt{5 - \sqrt{5}} + \\
& \quad \quad 36288 \sqrt{5 + \sqrt{5}} + 22335 \sqrt{2(5 + \sqrt{5})} - 20886 \sqrt{3(5 + \sqrt{5})} - 12941 \sqrt{6(5 + \sqrt{5})} \left. \right) \cos[x] + \\
& \quad 2 \left(-15 - 240 \sqrt{2} - 255 \sqrt{3} - 11397 \sqrt{5} + 325 \sqrt{6} - 1116 \sqrt{10} + 6927 \sqrt{15} + 399 \sqrt{30} - \right. \\
& \quad \quad 18644 \sqrt{30 - 6 \sqrt{5}} - 22146 \sqrt{15 - 3 \sqrt{5}} + 32496 \sqrt{10 - 2 \sqrt{5}} + 38070 \sqrt{5 - \sqrt{5}} - 36288 \\
& \quad \quad \left. \sqrt{5 + \sqrt{5}} - 22335 \sqrt{2(5 + \sqrt{5})} + 20886 \sqrt{3(5 + \sqrt{5})} + 12941 \sqrt{6(5 + \sqrt{5})} \right) \cos[2x] + \\
& \quad 250920 \sin[x] + 176040 \sqrt{2} \sin[x] - 147240 \sqrt{3} \sin[x] - 81924 \sqrt{5} \sin[x] - \\
& \quad 99960 \sqrt{6} \sin[x] - 59340 \sqrt{10} \sin[x] + 47180 \sqrt{15} \sin[x] + 34344 \sqrt{30} \sin[x] + \\
& \quad 53706 \sqrt{30 - 6 \sqrt{5}} \sin[x] + 80296 \sqrt{15 - 3 \sqrt{5}} \sin[x] - 95166 \sqrt{10 - 2 \sqrt{5}} \sin[x] - \\
& \quad 136044 \sqrt{5 - \sqrt{5}} \sin[x] + 175596 \sqrt{5 + \sqrt{5}} \sin[x] + 105486 \sqrt{2(5 + \sqrt{5})} \sin[x] - \\
& \quad 101528 \sqrt{3(5 + \sqrt{5})} \sin[x] - 60798 \sqrt{6(5 + \sqrt{5})} \sin[x] - 147270 \sin[2x] - \\
& \quad 102570 \sqrt{2} \sin[2x] + 81970 \sqrt{3} \sin[2x] + 49686 \sqrt{5} \sin[2x] + \\
& \quad 61380 \sqrt{6} \sin[2x] + 35490 \sqrt{10} \sin[2x] - 26930 \sqrt{15} \sin[2x] - \\
& \quad 21732 \sqrt{30} \sin[2x] - 29736 \sqrt{30 - 6 \sqrt{5}} \sin[2x] - 38628 \sqrt{15 - 3 \sqrt{5}} \sin[2x] + \\
& \quad 51084 \sqrt{10 - 2 \sqrt{5}} \sin[2x] + 67500 \sqrt{5 - \sqrt{5}} \sin[2x] - 82860 \sqrt{5 + \sqrt{5}} \sin[2x] - \\
& \quad 65142 \sqrt{2(5 + \sqrt{5})} \sin[2x] + 47392 \sqrt{3(5 + \sqrt{5})} \sin[2x] + 37926 \sqrt{6(5 + \sqrt{5})} \sin[2x] \left. \right) \Bigg) / \\
& \left(10455 + 7335 \sqrt{2} - 6135 \sqrt{3} - 3777 \sqrt{5} - 4165 \sqrt{6} - 2715 \sqrt{10} + \right. \\
& \quad 2105 \sqrt{15} + 1621 \sqrt{30} + \\
& \quad 1303 \sqrt{30 - 6 \sqrt{5}} + 2056 \sqrt{15 - 3 \sqrt{5}} - \\
& \quad 2433 \sqrt{10 - 2 \sqrt{5}} - 3312 \sqrt{5 - \sqrt{5}} + \\
& \quad 5538 \sqrt{5 + \sqrt{5}} + 3072 \sqrt{2(5 + \sqrt{5})} - \\
& \quad \left. 3194 \sqrt{3(5 + \sqrt{5})} - 1776 \sqrt{6(5 + \sqrt{5})} \right)
\end{aligned}$$

{True, True, True, True, True, True, True}



0.845182

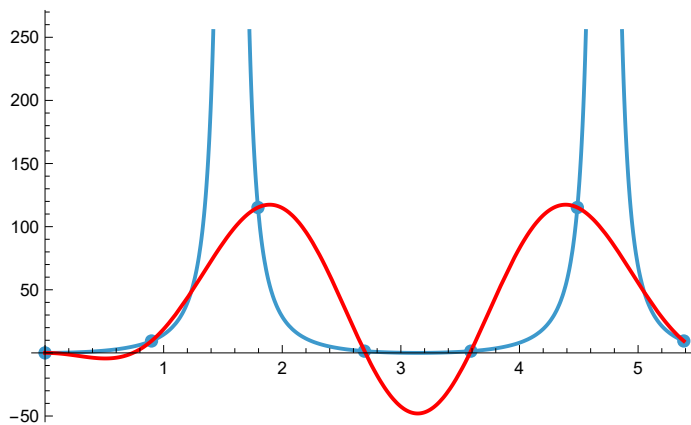
$$\begin{aligned}
& \left(2 \sin \left[\frac{\pi}{9} \right] \left(-91950 - 70935 \sqrt{2} + 55830 \sqrt{3} + 28878 \sqrt{5} + 39015 \sqrt{6} + 26901 \sqrt{10} - 18390 \sqrt{15} - \right. \right. \\
& 14317 \sqrt{30} - 18644 \sqrt{30 - 6 \sqrt{5}} - 22146 \sqrt{15 - 3 \sqrt{5}} + 32496 \sqrt{10 - 2 \sqrt{5}} + 38070 \sqrt{5 - \sqrt{5}} - \\
& 48864 \sqrt{5 + \sqrt{5}} - 42330 \sqrt{2(5 + \sqrt{5})} + 28632 \sqrt{3(5 + \sqrt{5})} + 24142 \sqrt{6(5 + \sqrt{5})} + \\
& 2(79410 + 53595 \sqrt{2} - 45180 \sqrt{3} - 32760 \sqrt{5} - 31415 \sqrt{6} - 17757 \sqrt{10} + 18774 \sqrt{15} + \\
& 10351 \sqrt{30} + 7443 \sqrt{30 - 6 \sqrt{5}} + 14400 \sqrt{15 - 3 \sqrt{5}} - 12501 \sqrt{10 - 2 \sqrt{5}} - 25494 \sqrt{5 - \sqrt{5}} + \\
& 36288 \sqrt{5 + \sqrt{5}} + 22335 \sqrt{2(5 + \sqrt{5})} - 20886 \sqrt{3(5 + \sqrt{5})} - 12941 \sqrt{6(5 + \sqrt{5})} \left. \right) \\
& \cos \left[\frac{\pi}{9} \right] + 2 \left(-15 - 240 \sqrt{2} - 255 \sqrt{3} - 11397 \sqrt{5} + 325 \sqrt{6} - 1116 \sqrt{10} + 6927 \sqrt{15} + \right. \\
& 399 \sqrt{30} - 18644 \sqrt{30 - 6 \sqrt{5}} - 22146 \sqrt{15 - 3 \sqrt{5}} + 32496 \sqrt{10 - 2 \sqrt{5}} + 38070 \sqrt{5 - \sqrt{5}} - \\
& 36288 \sqrt{5 + \sqrt{5}} - 22335 \sqrt{2(5 + \sqrt{5})} + 20886 \sqrt{3(5 + \sqrt{5})} + 12941 \sqrt{6(5 + \sqrt{5})} \left. \right) \\
& \cos \left[\frac{2\pi}{9} \right] + 250920 \sin \left[\frac{\pi}{9} \right] + 176040 \sqrt{2} \sin \left[\frac{\pi}{9} \right] - 147240 \sqrt{3} \sin \left[\frac{\pi}{9} \right] - \\
& 81924 \sqrt{5} \sin \left[\frac{\pi}{9} \right] - 99960 \sqrt{6} \sin \left[\frac{\pi}{9} \right] - 59340 \sqrt{10} \sin \left[\frac{\pi}{9} \right] + 47180 \sqrt{15} \sin \left[\frac{\pi}{9} \right] + \\
& 34344 \sqrt{30} \sin \left[\frac{\pi}{9} \right] + 53706 \sqrt{30 - 6 \sqrt{5}} \sin \left[\frac{\pi}{9} \right] + 80296 \sqrt{15 - 3 \sqrt{5}} \sin \left[\frac{\pi}{9} \right] - \\
& 95166 \sqrt{10 - 2 \sqrt{5}} \sin \left[\frac{\pi}{9} \right] - 136044 \sqrt{5 - \sqrt{5}} \sin \left[\frac{\pi}{9} \right] + 175596 \sqrt{5 + \sqrt{5}} \sin \left[\frac{\pi}{9} \right] + \\
& 105486 \sqrt{2(5 + \sqrt{5})} \sin \left[\frac{\pi}{9} \right] - 101528 \sqrt{3(5 + \sqrt{5})} \sin \left[\frac{\pi}{9} \right] - \\
& 60798 \sqrt{6(5 + \sqrt{5})} \sin \left[\frac{\pi}{9} \right] - 147270 \sin \left[\frac{2\pi}{9} \right] - 102570 \sqrt{2} \sin \left[\frac{2\pi}{9} \right] + \\
& 81970 \sqrt{3} \sin \left[\frac{2\pi}{9} \right] + 49686 \sqrt{5} \sin \left[\frac{2\pi}{9} \right] + 61380 \sqrt{6} \sin \left[\frac{2\pi}{9} \right] + \\
& 35490 \sqrt{10} \sin \left[\frac{2\pi}{9} \right] - 26930 \sqrt{15} \sin \left[\frac{2\pi}{9} \right] - 21732 \sqrt{30} \sin \left[\frac{2\pi}{9} \right] - \\
& 29736 \sqrt{30 - 6 \sqrt{5}} \sin \left[\frac{2\pi}{9} \right] - 38628 \sqrt{15 - 3 \sqrt{5}} \sin \left[\frac{2\pi}{9} \right] + 51084 \sqrt{10 - 2 \sqrt{5}} \sin \left[\frac{2\pi}{9} \right] + \\
& 67500 \sqrt{5 - \sqrt{5}} \sin \left[\frac{2\pi}{9} \right] - 82860 \sqrt{5 + \sqrt{5}} \sin \left[\frac{2\pi}{9} \right] - 65142 \sqrt{2(5 + \sqrt{5})} \sin \left[\frac{2\pi}{9} \right] + \\
& 47392 \sqrt{3(5 + \sqrt{5})} \sin \left[\frac{2\pi}{9} \right] + 37926 \sqrt{6(5 + \sqrt{5})} \sin \left[\frac{2\pi}{9} \right] \left. \right) \Bigg/ \\
& \left(10455 + 7335 \sqrt{2} - 6135 \sqrt{3} - 3777 \sqrt{5} - 4165 \sqrt{6} - 2715 \sqrt{10} + 2105 \sqrt{15} + \right. \\
& 1621 \sqrt{30} + 1303 \sqrt{30 - 6 \sqrt{5}} + 2056 \sqrt{15 - 3 \sqrt{5}} - \\
& 2433 \sqrt{10 - 2 \sqrt{5}} - 3312 \sqrt{5 - \sqrt{5}} + \\
& 5538 \sqrt{5 + \sqrt{5}} + 3072 \sqrt{2(5 + \sqrt{5})} - \\
& 3194 \sqrt{3(5 + \sqrt{5})} - 1776 \sqrt{6(5 + \sqrt{5})} \left. \right) - 6 \tan \left[\frac{\pi}{9} \right]^2
\end{aligned}$$

Задание 2

```

Clear[T, n, NN, X, F, koef];
f[x_] = 6 Tan[x]^2;
NN = 7;
X = Table[Subscript[x, k] = 2 Pi*k/NN, {k, 0, NN - 1}];
F = f[X];
Subscript[f, k_] := F[[k + 1]];
n = (NN - 1)/2;
Do[Subscript[C, k] = (1/NN)*Sum[Subscript[f, j]*Exp[I*2 Pi/NN]^(-k*j), {j, 0, NN - 1}], {
T[x_] = Sum[Subscript[C, k]*Exp[I*x]^k, {k, -n, n}] // FullSimplify
Table[Chop[T[Subscript[x, j]] - Subscript[f, j]] == 0, {j, 0, NN - 1}]
Gr1 = ListPlot[MapThread[List, {X, F}], PlotStyle -> {PointSize[0.02]}];
Gr2 = Plot[f[y], {y, Subscript[x, 0], Subscript[x, NN - 1]}];
Gr3 = Plot[T[y], {y, Subscript[x, 0], Subscript[x, NN - 1]}, PlotStyle -> Red];
Show[Gr1, Gr2, Gr3, PlotRange -> All]
-12 (Cos[x] + 5 Cos[2 x] - 3 (1 + Cos[3 x]))
{True, True, True, True, True, True, True}

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Задание 3

```

XX = {1/10, 2/10, 3/10, 4/10}; ff = {1.26, 2.864, 6.885, 9.764};
n = Length[XX] - 1; Subscript[xx, k_] := XX[[k + 1]]; Subscript[f, k_] := ff[[k + 1]];
eqv[m_, y_] := Table[(D[x^i, {x, m}] /. x -> y) == Sum[Subscript[d, k]*Subscript[xx, k]^i, {k, 0, n}], {m, 1, 4}];
Pr[m_, y_] := Sum[Subscript[d, k]*Subscript[f, k], {k, 0, n}] /. (Solve[eqv[m, y], {}]) //
{Pr[1, 1/10], Pr[1, 2/10], Pr[2, 0], Pr[3, 5/10]}
{-7.90833, 34.0567, 953.5, -3559.}

P[x_] = InterpolatingPolynomial[Table[{Subscript[xx, k], Subscript[f, k]}, {k, 0, n}], x];
Pr1[m_, y_] := D[P[x], {x, m}] /. x -> y;
{Pr1[1, 1/10] == Pr[1, 1/10], Pr1[1, 2/10] == Pr[1, 2/10], Pr1[2, 0] == Pr[2, 0], Pr1[3, 5/10] == Pr[3, 5/10]}
{True, True, True, True}

Pogr[m_, y_] := M/((n + 1)!) * Abs[D[Product[(x - Subscript[xx, k]), {k, 0, n}], {x, m}]] /.
{Pogr[1, 1/10], Pogr[1, 2/10], Pogr[2, 0], Pogr[3, 5/10]}
{M/4000, M/12000, 7 M/240, M/4}

```

```
Table[M/(n + 1)!*Maximize[{Abs[D[Product[(x - Subscript[xx, k]), {k, 0, n}], {x, m}]]],
{

$$\left\{ \frac{M}{10000}, \frac{M}{480}, \frac{7M}{240}, \frac{M}{4}, M, 0 \right\}$$

}
```